

Ordinary Differential Equations

Problem 1

Let $B(t)$ be the amount of money in a salesman's bank account at time t . Suppose that the salesman deposits money at a constant rate $d > 0$ and spends money at a rate proportional to the amount of money in the account. Find an ODE for $B(t)$.

Solution.

If the salesman never spent any money, the account would grow at a constant rate and $B(t)$ would satisfy the ODE

$$\frac{dB}{dt} = d.$$

On the other hand, if the salesman never deposited any money and only spent at a rate proportional to the amount of money in the account, the amount of money in the account would decrease exponentially and $B(t)$ would satisfy the ODE

$$\frac{dB}{dt} = -sB(t)$$

for some constant $s > 0$. If the salesman deposits money at a constant rate $d > 0$ and spends money at a rate proportional to the amount of money in the account, then $B(t)$ would satisfy the equation

$$\frac{dB}{dt} = d - sB(t)$$

for some positive constant $s > 0$. ■

Problem 2

Is $y = x^3$ a solution to the differential equation

$$xy' - 3y = 0?$$

Solution.

The power rule gives $y' = 3x^2$. It follows that

$$xy' - 3y = x \cdot 3x^2 - 3x^3 = 3x^3 - 3x^3 = 0,$$

so $y = x^3$ is indeed a solution to the differential equation $xy' - 3y = 0$.

Remark

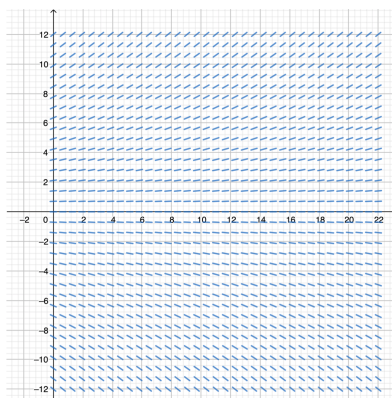
The differential equation $xy' - 3y = 0$ may be written as

$$\frac{dy}{dx} = \frac{3y}{x}$$

so it can be solved using separation of variables. This technique does indeed give the solution $y = x^3$. ■

Problem 3

Consider the following slope field.



Does Euler's method give an over-estimate or an under-estimate if

i) $y(0) = 2$?

ii) $y(0) = -2$?

Solution.

Euler's method gives an over-estimate if the solution curve is concave down and an under-estimate if the solution curve is concave up.

- i) If $y(0) = 2$, the solution curves up. In this case the solution curve is concave up so Euler's method gives an under-estimate.
- ii) If $y(0) = -2$, the solution curves down. In this case, the solution curve is concave down so Euler's method gives an over-estimate. ■

Problem 4

Consider the equation

$$\frac{dy}{dx} = y(1 - y).$$

Find all equilibrium solutions and sketch the slope field as well as some example solutions.

Solution.

The equilibrium solutions may be found by solving the equation $y(1 - y) = 0$ so they are $y = 0$ and $y = 1$. If $y > 1$, then

$$\frac{dy}{dx} = y(1 - y) < 0$$

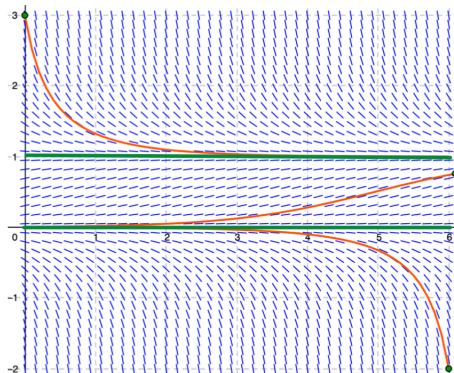
and the solutions decrease towards the equilibrium $y = 1$. If $0 < y < 1$, then

$$\frac{dy}{dx} = y(1 - y) > 0$$

and the solutions increase to the equilibrium $y = 1$. Finally, if $y < 0$, then

$$\frac{dy}{dx} = y(1 - y) < 0$$

and the solutions decrease to negative infinity. Notice that if $y = 1$ or $y = 0$ at any point in time, then $y = 1$ or $y = 0$ forever; this is the very definition of an equilibrium solution. This analysis allows us to sketch the slope field as well as some sample solutions.



Problem 5

Solve the differential equation

$$\frac{dy}{dx} = \frac{4x}{y^2},$$

and find the particular solution corresponding to the initial condition $y(0) = 1$.

Solution.

Cross multiplying and integrating gives

$$\int y^2 dy = \int 4x dx \quad \Rightarrow \quad \frac{y^3}{3} = 2x^2 + C.$$

It follows that $y^3 = 6x^2 + C$. Accounting for the initial condition shows that $1 = 6 \cdot 0 + C$. This means that $C = 1$ and $y^3 = 6x^2 + 1$. All in all,

$$y = \sqrt[3]{6x^2 + 1}.$$

Problem 6

Solve the differential equation

$$\frac{dy}{dt} = ky^2(1 + t^2).$$

Solution.

Cross multiplying and integrating gives

$$\int \frac{1}{y^2} dy = k \int (1 + t^2) dt \quad \Rightarrow \quad -\frac{1}{y} = k\left(t + \frac{t^3}{3}\right) + C = \frac{k(3t + t^3)}{3} + C.$$

Rearranging shows that

$$y = -\frac{3}{k(3t + t^3) + C}.$$

Problem 7

Solve the differential equation

$$\frac{dL}{dx} = k(x+a)(L-b).$$

Solution.

Cross multiplying and integrating gives

$$\int \frac{1}{L-b} dL = k \int (x+a) dx \quad \Rightarrow \quad \ln(L-b) = k\left(\frac{x^2}{2} + ax\right) + C.$$

Rearranging shows that

$$L = Ae^{k\left(\frac{x^2}{2} + ax\right)} + b.$$

■